Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/30/2007
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Project Title:	The Dynamics of Heterogeneous Solidification Microstructure Genesis in Undercooled Composite Systems		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	04/01/2003	**End Date:**	05/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 05/31/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNM04AA20G		
Performance Goal No.:			
Performance Goal Text:			

The ground based program represents a balanced experimental and analysis effort directed toward the investigation of drop tube, glass encased and levitation containerless processing methods. The investigation focus is on the understanding and analysis of microstructural evolution during solidification of undercooled melts and composite systems. The liquid undercooling level in a laboratory scale (3 m) drop tube and levitation melting system can be altered through the variation of processing parameters such as alloy composition, melt superheat, sample size and gas environment. The solidification behavior is evaluated through metallography, SEM, TEM, DTA, and x-ray diffraction in conjunction with a heat flow model of the processing conditions to judge the sample thermal history. A central component of the current program addresses the use of containerless processing together with an undercooled melt with incorporated particles to develop a critical evaluation of solidification front-particle interactions. Another component of the current investigation

Task Description:	is the analysis of the control of microstructural evolution in Ni-V, Nb-Pd and Co-Al alloys through the high undercooling levels provided by containerless processing to provide fundamental new information on the control of nucleation. The microgravity environment provides new opportunities for the identification and characterization of key parameters that control of solidification processes, and the containerless environment afforded by microgravity conditions avoids crucible-induced sample contamination and allows for large undercooling. The main research objective is the evaluation and analysis of the undercooling and resultant solidification microstructures in containerless processing, including drop tube processing and levitation melt processing of selected alloys and composites. Containerless processing in ground-based drop tubes simulates microgravity conditions via solidification of liquid samples under free fall conditions. The containerless environment is also attained in levitation melt processing and removes a major source of impurities and heterogeneous nucleation sites, allowing for a large melt undercooling. This enhanced liquid undercooling exposes alternate solidification pathways, allowing for the formation of novel microstructures. Controlling the undercooling level provides some control of the operative solidification pathway and the resultant microstructure. The novel structures that may be produced in a ground-based containerless processing facility preview the wide range of possible materials processing experiments that may be conducted in a space-based laboratory. The results of the ground-based study will be used to identify critical experimental variables in microgravity processing and the analysis can be used to design and define the science and hardware requirements for extended duration space experiments. In the solidification of composites the interaction between the solidification front and particles is central to effective processing. Undercooled liquid solidification simulates convection-free solidification conditions for such composites in a 1g environment. Numerous models for the interaction behavior have been proposed for planar solidification fronts, but critical tests are lacking in many cases due to confounding gravitational effects. The interaction of a fast-moving, dendritic solidification front with dispersed particles during solidification processing of metal-matrix composites has revealed a new mode of the incorporation behavior that is not included in existing models for planar interfaces and has not been experimentally accessed previously. In order to complete the analysis of the new microstructural class developed during duplex partitionless solidification it is important to address the key issue of independent or coordinated nucleation of the product phases. Experiments on undercooled samples have provided the microstructural characteristics that are essential to answer this issue. With the critical information on the nucleation mechanism a complete kinetics model analysis can be formulated for the duplex partitionless reaction. The outcome of this program will provide a unique background for a space experiment, because never before has it been possible to quantify the solidification kinetics and the associated microstructure evolution in a manner that can be achieved with the proposed experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Basic understanding of composite materials manufacturing
Task Progress:	The current studies have established that some pure metals and alloys can exhibit large undercooling levels in the presence of crystalline particles. A wide range of growth velocities can be examined through variation of the undercooling levels. Highly undercooled samples provide an excellent means to study the interaction of high-velocity dendrites and particles. The first set of experiments was focused on the examination of the transitions between particle incorporation modes (i.e. pushing, entrapment, and engulfment). Another set of experiment was focused on the accessible undercooling and the resultant solidification microstructure of combinations of pure metals with a dispersion of crystalline particles of various diameters with respect to the process of particle incorporation. The studies have shown that the transition from entrapment to engulfment was not observed in the copper-aluminum oxide, copper-zirconium oxide, and nickel-zirconium oxide composite system at any experimentally accessible undercooling (up to 120 K). Samples with copper-nickel alloy matrix materials and zirconium oxide particles at similar undercooling levels (up to 160 K) also do not exhibit engulfment. The transition from entrapment to engulfment was observed in copper-tantalum oxide and nickel-tantalum oxide composite systems at the growth velocity that corresponds to an achieved undercooling of 173 K. Therefore, these two composite systems provide an excellent means to study the interaction of high velocity dendrite and particles and test theoretical models over an extended range of dendrite tip velocities. In addition, these composite systems can be used to simulate buoyancy-neutral experimental conditions in a terrestrial setting. The studies have shown that large undercooling levels can be achieved in copper-tantalum oxide and nickel-tantalum oxide composite systems and these undercooling levels depend on the particle size. Within the achievable undercooling range of nickel-tantalum oxide composite system, the microstructure evolution with undercooling was investigated. The particle sizes were found to alter the mechanism of grain refinement in an undercooled melt. The grain refinement occurring at high undercooling levels can be ascribed to the breakup of the dendrite-skeleton or the splitting of the dendrite arms owing to the high stress interaction of dendrites with large particles originating from rapid contraction upon solidification. Grain refinement is also possibly initiated by the fluid flow due to convection. Flux treatments with a molten glass slag are commonly used to promote deep liquid undercooling or even bulk glass formation. Although the technique has demonstrated frequently the capability to yield large undercooling, the underlying mechanisms remain uncertain. There are suggestions of flux-induced “ nucleant removal” or “surface site deactivation”, but the effectiveness and operations of these processes have not been identified clearly. As a model system, the undercooling response of pure gold encased in Pyrex glass was studied systematically to develop a consistent record of the undercooling behavior. Nucleation kinetics analyses of statistically significant sets of measurements performed under strictly controlled conditions reveal the presence of a new mechanism based on gas-solid interactions that trigger nucleation, through a nucleant precipitation as the first step of the interaction. A new model based on thermodynamic considerations and on the nucleation kinetics data are proposed, which accounts for the undercooling increase including initial conditioning, atmosphere effects and undercooling saturation in a self-consistent manner. The current studies will provide a database for future space experiments where the incorporation behavior can be examined in the absence of the confounding influence of buoyancy-induced convection. The combination of ground-based studies and experiments under microgravity conditions will provide an understanding of the key parameters of the particle incorporation process that can also lead to improved composite processing strategies for both uniform gradient particle distributions.

Bibliography Type:	Description: (Last Updated: 04/29/2010)
Articles in Peer-reviewed Journals	Wilde G, Sebright JL, Perepezko JH. "Bulk liquid undercooling and nucleation in gold." Acta Materialia, 2006 Oct; 54(18):4759-69. http://dx.doi.org/10.1016/j.actamat.2006.06.007 , Oct-2006
Dissertations and Theses	Santhaweesuk C. "Particle-dendrite interactions and microstructural evolutions during the solidification of undercooled metal-matrix composites." University of Wisconsin-Madison, May 2006. , May-2006